

Sensors

Welcome To The INTERNET OF NANOTHINGS



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Sensors are everywhere; in your mobile phones, laptops, security devices and vehicles. Sensors like accelerometers, gyroscopes, temperature sensors, acoustic sensors and flow sensors that are already available in the market are (relatively) too large in size for some of the cutting-edge applications that are waiting for these.

"The sensors are available, but you need to miniaturise these a lot more. Instead of the Internet of Things (IoT), I would call it the Internet of Nanothings, because until you miniaturise the sensors and make these cost-effective, developers will not be able to utilise these," said Dr Vijay Mishra, chief technology officer at Centre for Nano Science and Engineering (CeNSE) at Indian Institute of Science (IISc), Bengaluru, in a session delivered at India Electronics Week 2016.

Autarky in sensors

Autarky means economic independence or self-sufficiency. In sensor talk, energy autarkic sensors are those that can obtain energy needed from ambient temperature differences in the environment. Why? These temperature differences let the devices operate without batteries or wiring, thus enabling

these to be self-sufficient little pieces of technology that can survive on their own.

Fraunhofer Institute of Physical Measurement (IPM) develops thermoelectric wireless sensors that can generate enough energy to supply various sensors, including their radio units, with enough energy to function properly. These can be used to implement wireless monitoring of body functions or to monitor material fatigue in aircraft.

The average German drinks about 150 litres of coffee per year, claims Fraunhofer IPM. Unsurprisingly, one of the popular innovations from energy-autarkic sensors led to the development of the communicating coffee pot. The pot uses heat from hot coffee to power the high-tech coffee pot's sensor technology, which keeps a watch on coffee temperature and lets the kitchen team know if its coffee has gone cold. Ingenious, would you not say?

Wristbands and kits

Other examples of energy harvesting include BlueTEG sensor wristband that uses the temperature difference between the skin and the surrounding air to produce energy, and transmits data via Bluetooth. Sensors in applications to test structural integrity use piezoelectric collectors to gather energy from vibrational stresses.

One among the numerous solutions available is Wurth Electronics' Energy Harvesting Solution To Go Kit, which helps developers apply these technologies in battery-less products. The kit can harvest energy through solar, electromagnetic, piezoelectric and thermal sources.

Sensing in biomedical

There are some interesting sensors developed at CeNSE such as Envirobat, a gas sensor. Others include pressure sensors ranging from those that can be used in the aeronautical industry that can measure 500

